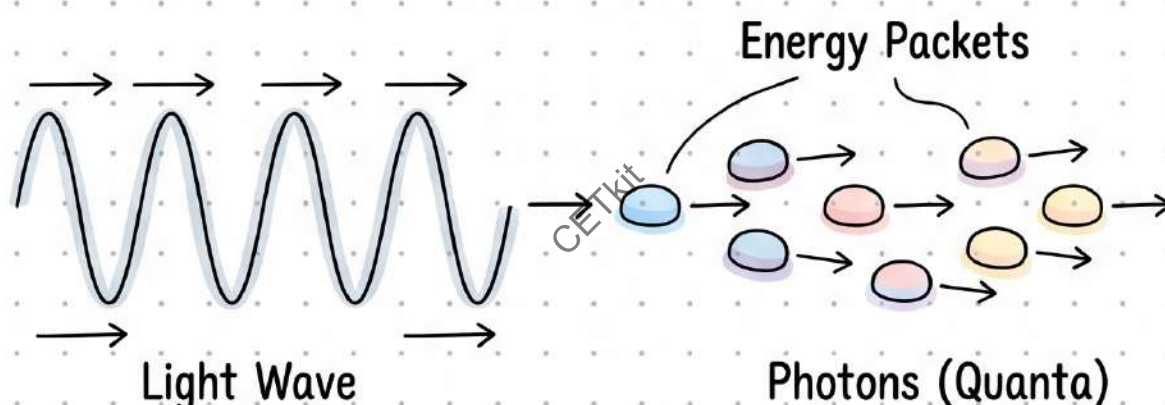


Dual Nature: Photons & Quantum Theory

Planck's Quantum Theory

Planck's Quantum Theory

- According to Planck's quantum theory, light is considered to be made up of small packets (or particles) of energy known as quanta of energy or radiation. Einstein proposed that electromagnetic radiation (or simply light) is quantized and exists in elementary amounts (quanta) that we now call photons.



Properties of Photons

- Energy: $E = h\nu = \frac{hc}{\lambda} = \frac{12400}{\lambda(\text{\AA})} \text{ eV}.$
- Momentum of one photon is $\frac{h}{\lambda}.$
- Photons are not deflected by electric and magnetic fields which shows that they are neutral and do not carry any charge.
- The energy of photon depends upon the frequency of radiation but is independent of the intensity of radiation.

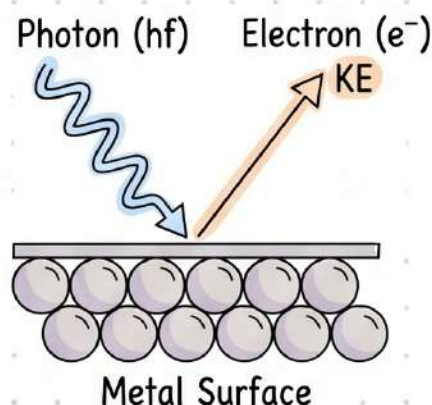
Photoelectric Effect Fundamentals

Photoelectric Emission:

When light of suitable frequency illuminates a metal surface, electrons are emitted.

This process of ejection of electrons using light is known as photoelectric emission.

Photoelectrons ejected from metal have kinetic energies ranging from 0 to $K_{\max}$.



Work Function (ϕ):

A certain minimum amount of energy is required for an electron to be pulled out from the surface of a metal. This minimum energy is called the work function (ϕ) of that metal. Work function is minimum for cesium (1.9 eV).

Threshold Parameters:

The minimum energy is frequency of light below which

$$hf = h\nu_0$$

$$\phi = h\nu = hc/\lambda$$

$$\lambda_0 = \frac{hc}{\phi}$$

Threshold Parameters:

The minimum frequency of the incident light below which photoelectrons are not ejected from the metal surface is known as threshold frequency (ν_0).

Work function, $\phi = h\nu_0 = \frac{hc}{\lambda_0}$, where λ_0 is the threshold wavelength.

Einstein's Equation & Stopping Potential

Einstein's Photoelectric Equation:

$$h\nu = \phi + KE_{\max} \Rightarrow \frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV_s$$

$h\nu$: Incident photon energy
 ϕ : Work function
 $KE_{\max}$: Max kinetic energy of ejected electron

$$\text{Work function, } \phi = h\nu_0 = \frac{hc}{\lambda_0}$$

Stopping Potential (V_s)

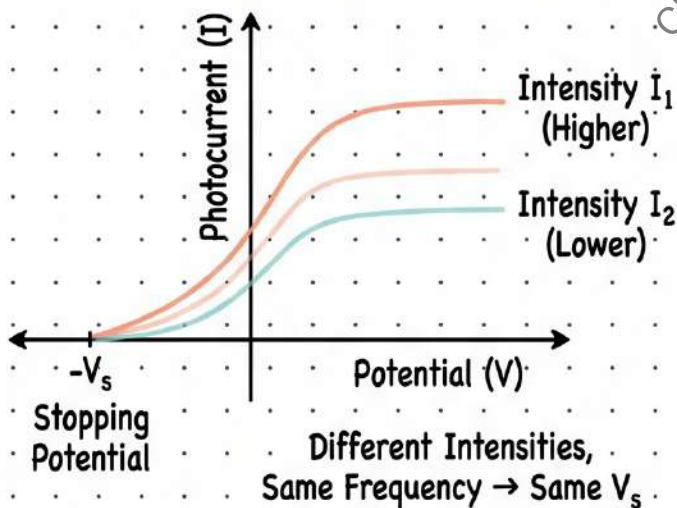
The minimum negative potential applied to the metal plate with respect to the collector, at which the photoelectric current becomes zero.

Here, $KE_{\max} = eV_s$, where V_s is the stopping potential.

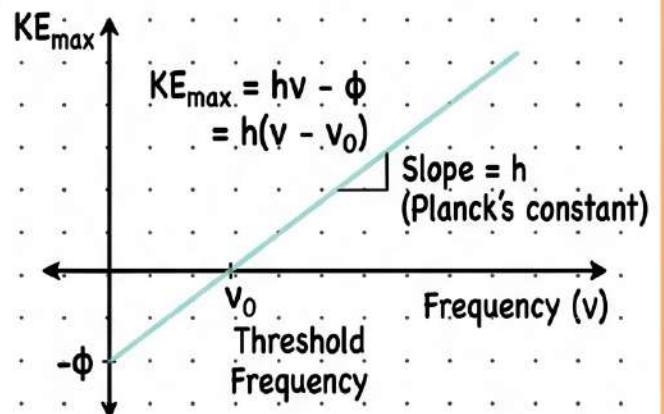
Note: Stopping potential is independent of the intensity of incident light.

Stopping Potential V_s depends only on the FREQUENCY (or wavelength) of incident light and the NATURE of the metal surface.

Photocurrent (I) vs. Potential (V)



$KE_{\max}$ vs. Frequency (ν)



Linear relationship. No emission if $\nu < \nu_0$. Slope is independent of the metal.

Laws of Photoelectric Emission

- The number of photoelectrons emitted per second is directly proportional to the intensity of incident radiation.
- The maximum kinetic energy ($KE_{\max}$) of the ejected electrons is independent of the intensity of incident radiation.
- $KE_{\max}$ depends linearly on the frequency of the incident radiation (above threshold ν_0).
- Photoelectric emission is an instantaneous process.

de-Broglie Waves & Heisenberg

• Radiation Pressure: $P = \frac{I}{c} (1+r)$

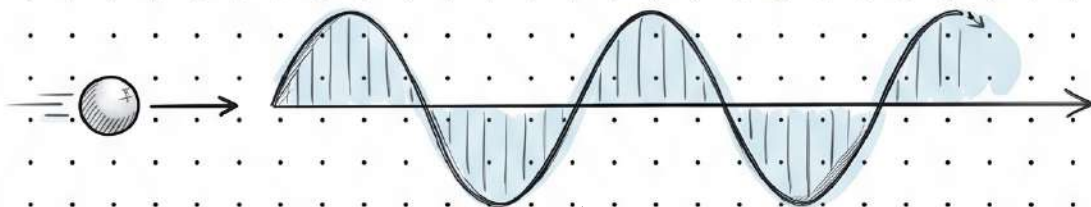
↳ (where I is intensity, c is speed of light, and r is reflectivity).

For 100% reflection, $r = 1$

For 100% absorption, $r = 0$

• de-Broglie Hypothesis:

A wave is associated with a moving material particle.



de-Broglie wavelength of a material particle: $\lambda = \frac{h}{mv}$

In terms of energy of a particle (E): $\lambda = \frac{h}{\sqrt{2mE}}$

Of an electron accelerated through a potential V volt: $\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$
(V in Volts)

In terms of temperature (T): $\lambda = \frac{h}{\sqrt{3mkT}}$

• Heisenberg's Uncertainty Principle:

It is not possible to measure exactly both the position and momentum of a microscopic particle (say electron) at the same time.

$$\Delta x \Delta p \geq \frac{\hbar}{2}, \text{ where } \hbar = \frac{h}{2\pi}$$